

Ali Ayan

Contact Details: +491638041149 | aliayanlakhani@gmail.com

Address: Forst-Kasten-Allee 131, München 81475

LinkedIn: <https://www.linkedin.com/in/ali-ayan-lakhani/>

Date of Birth: 19.07.2000



Relevant Experience

Krones AG | Working Student – AI / Computer Vision

Raubling, Bavaria, Germany | Oct 2025 – Present



- Automated Detection and Localization of Industrial Equipment in Engineering Layouts
- Developed Rotation-Invariant Computer Vision Pipelines for CAD and Factory Drawings
- Achieved ~80% Automation of Layout Generation through Combined Classical and Deep Learning Methods
- **Skills:** Computer Vision, Deep Learning, Feature Extraction, Object Localization, Human-in-the-Loop AI Systems
- **Software:** Python, OpenCV, YOLO, Detectron2, NCC, Canny Edge Detection

AUDI AG | Bachelor Thesis

Neckarsulm, Germany | Oct 2024 – Apr 2025



- Uncertainty Quantification for Automotive Crash Boxes: A Monte Carlo Simulation Approach
- **Skills:** Framework development, automotive structural analysis
- **Software:** PamCrash, Python, Bash, LoCo (Loadcase Composer developed by SCALE)

AUDI AG | CAE Structural Load Simulation Engineering Intern

Neckarsulm, Germany | Apr 2024 – Oct 2024



- Conducted validation experiments and parametric studies on CMS structures, improving simulation accuracy through material property analysis.
- Contributed to the design and optimization of body structures for Audi A4/A5 (B10) and Audi A7 (C9), ensuring structural integrity and safety.
- **Skills:** Validation experiment setup, simulation model optimization, material sensitivity analysis, CAD-based structural design, collaborative R&D
- **Software:** ANSA, Animator4, PamCrash, CATIA

AUDI AG | Audi Hackathon 2023

Apr 2023



- Developed AI-driven solutions to enhance internal processes, using Java Object Oriented Programming (OOP) and Large Language Models (LLM).
- **Skills:** AI integration, Java OOP, LLM deployment, cross-functional teamwork

European Young Engineers (EYE) | Head of the project HorEYEzon

June 2022 – Present



- Facilitated dialogues on global engineering topics with key stakeholders across Europe, enhancing cross-cultural collaboration within EYE's network.
- **Skills:** Stakeholder engagement, intercultural project leadership, public speaking, strategic networking

The Hague University of Applied Sciences | EXPLICO Project

Nov 2022 – Dec 2022



- Developed solutions for local issues within multicultural virtual exchange teams, fostering cross-cultural adaptability.
- **Skills:** Virtual collaboration, cross-cultural adaptability, problem-solving in diverse teams

Education



Technical University of Munich
Master of Science (M.Sc.) Computational Mechanics

Munich, Germany | Present



Hochschule Rhein-Waal
B.Sc Mechatronics Systems Engineering – GPA: 2.0 (German Grading System)
Focus Field: Simulation in Mechatronics

Kleve, Germany | Oct 2020 – May 2025



The Westminster School
Primary and Secondary schooling – GPA: 2.0 (German Grading System)
International General Certificate of Secondary Education (IGCSE) & Advanced Levels (A-levels)

Dubai, United Arab Emirates | Oct 2009 – Oct 2019



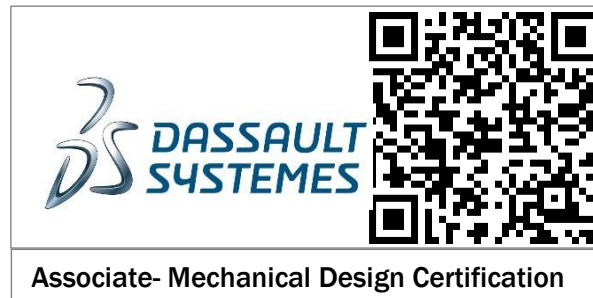
Additional Projects

Autonomous Track Following Robot	IEEE ADR Project
<i>Kleve, Germany Apr 2023 – Jun 2023</i> - Basic experience with ROS and OpenCV - Autonomous track following using line detection - Gained intermediate experience in with Python and NumPy - Object tracking via front-camera on robot, using Haar Cascade libraries	<i>Kleve, Germany Mar 2021 – Aug 2021</i> - ADR – Autonomous Delivery Robot - Role: Embedded System Design - Experience with Raspberry Pi and Arduino, programming in Arduino IDE - Gained an understanding of LiDar technology and the basics of SLAM
Temperature Sensor and Logger	Automatic Sprinkler System
<i>Kleve, Germany Sep 2021 – Jan 2022</i> - Experience with ATMEL microcontrollers and EEPROM, temperature sensors and LCD modules - Programming in C, using Notepad++ - Implemented and tested with different communication buses such as I2C, SPI and UART	<i>Kleve, Germany Oct 2022 – Jan 2022</i> - Design of sprinkler and housing on SolidWorks - Manufacturing drawings & circuit layouts - Project management, DFMEA

Competencies

Technical Skills

Altium, Autodesk EAGLE	Basic
Analog and Digital Electronics design	Basic
Ansys Mechanical (FEM)	Intermediate
C/C++	Intermediate
Object Oriented Programming in Java	Intermediate
Computer-aided-design (CAD)	Advanced
Pre & Post Processing using ANSA and Animator4	Advanced
Circuit simulations (LTspice)	Advanced



Languages

- Arabic – Intermediate (A2)
- English – Advanced (C1)
- German – Intermediate (A2)
- Urdu – Proficient (C2)

Social Skills

- Strong communication and negotiation skills
- Conflict management
- Self-starting attitude